NOT MEASUREMENT SENSITIVE

MIL-F-3222F
16 July 1991
SUPERSEDING
MIL-F-3222E
26 October 1978

MILITARY SPECIFICATION

FLOODLIGHTS AND LANTERNS, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for methods, materials, and procedures, for the preservation, packing, and marking of floodlights and lanterns (except trailer-mounted floodlights) (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-P-40	-	Packaging and Packing of Hand Tools.
PPP-T-60	-	Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A AREA PACK

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MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.

STANDARDS

FEDERAL

FED-STD-101	-	Preservation, Packaging, and Packing Materials: Test Procedures.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953	-	Strapping, Flat, Steel and Seals.
D 4675	-	Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this

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document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials. Materials shall be as specified herein and in applicable specifications and standards. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects (see 6.5).

3.3 Preservation. Preservation shall be level A or C as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts vulnerable to damage.

3.3.1.2 Methods. Methods of preservation indicated by symbols herein shall conform to the corresponding symbols in MIL-P-116.

3.3.1.3 Preservatives. Preservatives shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

3.3.1.4 Unprotected metal surfaces. Preservatives shall not be applied to reflectors. Exterior unprotected metal surfaces of the floodlights, lanterns, and accessories, requiring the application of a contact preservative in accordance with the preservative application criteria of MIL-P-116, shall be coated with type P-19 preservative. Interior surfaces, machined surfaces, and threaded surfaces, if unprotected, and requiring the application of a contact preservative in accordance with the preservative application criteria of MIL-P-116, shall be coated with type P-7 preservative. Excess preservative shall be allowed to drain. Exposed surfaces coated with type P-7 preservative shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, type II, grade A, class 2, and the wrap or cover secured in place with tape conforming to PPP-T-60, type IV.

3.3.1.5 Repair parts and accessories. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts and accessories. When sufficient space is available, the preserved repair parts and accessories shall be placed within the container with the end item for which they are intended. When sufficient space is not available for placing with the end item, the preserved repair parts and accessories shall be consolidated as specified in 3.3.1.11.

3.3.1.6 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, level A. When sufficient space is available, the

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preserved tools shall be placed within the container containing the end item for which they are intended. When sufficient space is not available for placing with the end item, the preserved tools shall be consolidated as specified in 3.3.1.11.

3.3.1.7 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1. When space is available the preserved publications shall be placed within the container with the end item. When space is not available for placing with the end item, the preserved publications shall be consolidated as specified in 3.3.1.11.

3.3.1.8 Electric operated and battery operated floodlights and lanterns.

3.3.1.8.1 Enclosed, gasketed, and weatherproofed. Each complete floodlight and each complete lantern shall be preserved in accordance with MIL-P-116, method III. Containers shall conform to PPP-B-636, W5c, W6c, or V3c, as applicable, style RSC. Cushioning shall be provided to immobilize the items within the containers. Box closure shall be in accordance with the appendix to the box specification, method V.

3.3.1.8.2 Not enclosed, not gasketed, and not weatherproofed. Each complete floodlight and each complete lantern shall be preserved in accordance with MIL-P-116, method IA-14. When a rigid carrying case is furnished, it may be used as the inner container. Cushioning shall be provided within the inner containers and in rigid carrying cases to immobilize the items within the containers. The outer containers shall conform to PPP-B-636, W5c, W6c, or V3c, as applicable, style RSC. Box closure shall be in accordance with the appendix to the box specification, method V.

3.3.1.9 Carbide, acetylene, and other gas-illuminated floodlights and lanterns. Each complete floodlight and each complete lantern shall be preserved in accordance with MIL-P-116, method IC-2.

3.3.1.10 Gasoline and oil-burning floodlights and lanterns. Removed mantles and replacement mantles for each item shall be preserved in a kraft envelope and the envelope sealed. Each complete floodlight and each complete lantern, with preserved mantles, shall be preserved in accordance with MIL-P-116, method IC-2.

3.3.1.11 Consolidation. Repair parts, tools, technical publications, and accessories such as mounting masts, tripod mountings, and globes for each floodlight or lantern that is not preserved in the container with the end item, shall be consolidated in a close-fitting box conforming to PPP-B-636, W5c or W6c, style RSC. Cushioning shall be provided in accordance with MIL-STD-1186 to immobilize the items within the boxes. Box closure shall be in accordance with the appendix to the box specification, method V.

3.3.2 Level C. The floodlights and lanterns and all accessories shall be preserved in accordance with the contractor's standard practice in a manner to

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prevent deterioration and damage during shipment from the contractor to initial destination.

3.4 Packing. Packing shall be level A or C as specified (see 6.2).

3.4.1 Level A. Complete floodlights and lanterns of like description preserved as specified in 3.3, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style I or J. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

3.4.2 Level C. The floodlights and lanterns preserved as specified in 3.3, shall be prepared for shipment in a manner which will ensure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

3.5 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

3.6 Workmanship. Cleaning, preserving, packaging, packing and marking requirements specified herein shall give maximum protection against deterioration, and mechanical or physical damage, during handling, shipment, and storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Material inspection. The contractor is responsible for insuring that materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article pack inspection.

4.3.1 Examination. The first article pack shall be examined as specified in 4.4.4. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Unless otherwise specified (see 6.2), the first article pack for level A and C shall be tested in accordance with 4.3.2.1 or 4.3.2.2 as applicable to the gross weight of the box. Failure of the applicable test shall be cause for rejection.

4.3.2.1 Packs not exceeding 200 pounds gross weight. The pack shall be subjected to the free-fall drop test in accordance with FED-STD-101, test method 5007.

4.3.2.2 Packs exceeding 200 pounds gross weight. The pack shall be subjected to the pendulum-impact test in accordance with FED-STD-101, method 5012.

4.3.3 Basis for rejection. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing or any visual damage to the contents shall constitute failure of the tests.

4.4 Quality conformance inspection.

4.4.1 Inspection stages. Inspection shall be in two stages as follows:

a. The first stage shall include inspection of materials, methods, containers, and marking within the shipping container.
b. The second stage shall include exterior inspection of materials and marking after closing the shipping container.

4.4.2 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Samples shall be selected using table I and IIa of MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level of packaging as specified in table I. Presence of one or more defects shall be cause for rejection.

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TABLE I. Classification of defects.

No.	Defect	Level A	Level C
101.	Materials, methods, or containers not as specified. Each incorrect material, method, or container shall constitute one defect (see 3.2, 3.3, and 3.4).	X	-
102.	Surfaces not coated with preservatives as specified (see 3.3.1.4).	X	-
103.	Coated surfaces not wrapped or covered as specified (see 3.3.1.4).	X	-
104.	Box closure not method V as specified (see 3.3.1.8.1, 3.3.1.8.2, and 3.3.1.11).	X	-
105.	Replacement mantels and removed mantles not preserved as specified (see 3.3.1.10 and 3.3.2).	X	X
106.	Consolidation not as specified (see 3.3.1.11).	X	-
107.	Cushioning not provided as specified (see 3.3.1.11).	X	X
108.	Floodlights and lanterns not preserved as specified (see 3.3.2).	-	X
109.	Strapping not as specified (see 3.4.1).	X	-
110.	Floodlights and lanterns not packed as specified for level C (see 3.4.2).	-	X
111.	Marking illegible, missing, incomplete, or incorrect (see 3.5).	X	X
112.	Workmanship not as specified (see 3.6).	X	X

5. PACKAGING

(This section not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is intended to be used for reference in section 5 of commodity specifications for floodlights and lanterns (except trailer-mounted floodlights) or for direct reference in contracts, and as an aid in the preparation of packaging data sheets.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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c. When a first article pack is not required (see 3.1 and 4.3.2).
d. Level of preservation and packing required (see 3.3 and 3.4).
e. Any special marking (see 3.5).

6.3 Trailer-mounted units. Preservation, packing, and marking of trailer-mounted floodlights will be in accordance with the packaging requirements of the applicable end item specification.

6.4 First article pack. The first article pack may be accomplished utilizing either the preproduction model or a production floodlight or lantern. If the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and the production floodlights or lanterns. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the floodlights and lanterns in accordance with this specification.

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Subject term (key word) listing.

Marking
Packing
Preservation

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Army - SM
Air Force - 82

User activity:
Navy - MC

Preparing activity:
Army - ME

Project PACK-0930

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INSTRUCTIONS

1. The preparing activity must complete blocks 1,2,3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4,5,6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of this letter.

NOTE: This form may not be used to request copies of documents, nor to request waives requirements in current contracts. Comments submitted on this form do not constitute or imply autdhorization to waive any portion of the referenced document(s) or tdo amend contractural requirements.

RECOMMENDED CHANGE:	1. DOCUMENT NUMBER MIL-F-3222F	2. DOCUMENT DATE (YYMMDD) 910716

3. DOCUMENT TITLE
Floodlights and Lanterns, Packaging Of

4. NATURE OF CHANGE (identify paragraph number and include proposed if possible. Attach extra sheets as needed).

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (LAST,FIRST MIDDLE INITIAL)	b. ORGANIZATION	
c. ADDRESS (Include zip code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Name Carolyn B. Johnson	b. TELEPHONE (In lude Area Code) (1) Commercial (2) AUTOVON (703) 664-5717 354-5717
ADDRESS (Include zip code) Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	If you do not receive a reply with 45 days, contact: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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